

DIVYANSHU AGRAHARI

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Education

Thapar Institute of Engineering and Technology, Patiala

B.E. Computer Engineering (Upgraded from Electronics & Communication Engineering)

CGPA: 9.18/10

Sept 2021 – June 2025

Professional Experience

MITACS Globalink Research Intern – University of Calgary

Intelligent Dynamics & Control Lab (IDCL)

Summer 2024

Supervisor: Dr. Mahdis Bisheban

Python, ROS 2, Nav2, MoveIt 2, ros2_control, LIO-SAM, slam_toolbox, Gazebo

- Extended a **ROS 2** / Gazebo aerial manipulator (quadcopter with a 7-DoF Kinova Gen3 arm, MoveIt 2, ros2_control and an SE(3) geometric controller) with **autonomous navigation and SLAM**.
- Integrated the Nav2 navigation stack with the geometric flight controller through a **Python trajectory layer** that converts Nav2 velocity commands into smooth trajectories, enabling **fully autonomous goal-to-goal flight**.
- Deployed **3D LiDAR–inertial SLAM** (LIO-SAM + IMU fusion) and 2D SLAM (slam_toolbox), tuning parameters and launch files for accurate **GPS-denied mapping and localization**.
- Built a **reusable simulation testbed** with a single-config launch system (UAV-only vs UAV+arm, 2D vs 3D SLAM), validated across custom GPS-denied mazes, with real-time plotting and threaded logging to debug trajectory tracking.

Projects

VibeCraft – AI project builder with live previews

🐙 [GitHub](#) 🌐 [Live Demo](#)

Java 25, Spring Boot 4.1, Spring Cloud, Spring AI, PostgreSQL, Flyway, Redis, MinIO, React 18, TypeScript, Kubernetes (k3s), GitHub Actions

- Architected as **Spring Boot microservices** behind a **Spring Cloud Gateway** with Eureka, **database-per-service** PostgreSQL (Flyway) and a secret-authenticated OpenFeign internal API; an AI-written interview turns a one-line idea into a spec.
- Streamed **LLM code generation** with **Spring AI** (OpenRouter) in a tag-based protocol with a live build checklist and auto-recovery, publishing each AI turn as one **atomic, restorable revision** to MinIO object storage.
- Ran every project's live preview in an **isolated Kubernetes pod** claimed from a **warm pool** and reclaimed when idle, served through **signed, expiring links** routed by Redis and a Node.js reverse proxy.
- Built the **React + TypeScript** SPA (Vite, TanStack Query, CodeMirror) with Firebase Auth, **role-based access** per project and Stripe-metered usage limits.
- Tested with JUnit and **Testcontainers**, and shipped through a **GitHub Actions CI/CD** pipeline that builds arm64 images, deploys to k3s on Oracle Cloud behind Cloudflare Tunnel, smoke-tests and **auto-rolls back**.

PayFlo – payment gateway backend

🐙 [GitHub](#)

Java 25, Spring Boot 4.1, Spring Cloud 2025.1, Apache Kafka, Redis, PostgreSQL, Resilience4j, Spring Data JPA, Docker, Jib, Kubernetes (kind)

- Designed the gateway as **Spring Boot microservices** (merchants, orders, payments, card vault, webhooks, payouts) behind a single API gateway with service discovery and centralized config, using **circuit breakers** and **idempotent writes** so a retried or failed call never double-charges a customer.
- Authenticated staff (**JWT**) and merchant backends (API keys) at the gateway with **Redis rate limiting**; routed card, UPI and net-banking payments through a payment state machine and a **saga** that never holds a DB transaction across a network call.
- Published every domain change to **Kafka via a transactional outbox** so no event is lost to a crash, and delivered **HMAC-signed webhooks** retried on a fixed schedule then dead-lettered, with ShedLock keeping scheduled jobs single-run across replicas.
- Isolated **PCI-scoped card data** in a vault service with per-card **AES-256-GCM envelope encryption** (merchants see only tokens), ran nightly settlement with fee and GST breakdown, and deployed on **Kubernetes** with Kustomize and Jib.

ThaparGenie – RAG assistant for Thapar Institute students

🐙 [GitHub](#) 🌐 [Live Demo](#)

Python, Django 5.2, DRF, PostgreSQL + pgvector (Supabase), Gemini API, React 19, TanStack Query, Tailwind CSS v4, Firebase Auth, Playwright

- Built a **citation-grounded RAG assistant** over 12,600+ passages from 1,300+ official TIET documents: questions are rewritten into standalone queries, retrieved by **hybrid search** (pgvector HNSW + Postgres full-text, fused with RRF), and a **grounding check** flags claims not backed by the cited sources.
- Modelled conversations as a **message tree** so edits and regenerations branch instead of overwriting history, with answers **streamed over SSE** (stop and resume) and a **semantic answer cache** cleared whenever the knowledge base changes.
- Built an admin dashboard and a **crash-recoverable ingestion pipeline** that turns PDFs, Word and Excel files, web pages and FAQs into searchable passages, with validity dates, notice sync and a nightly **recall@5 check** that warns when search quality drops.
- Secured it for public use with Firebase Auth and approval-based access, per-user and global AI-call limits, **row-level security** on every table, an audit log and **DPDP Act-aligned retention**, with **Playwright end-to-end tests** in CI.

Next.js 16, React 19, TypeScript, Prisma 7, PostgreSQL (Neon), NextAuth v5, Tailwind CSS v4, shadcn/ui, Zod, Chrome Extension, Vitest, Vercel

- Built a **full-stack Next.js (App Router)** app with no separate API server: **React Server Components** read through **Prisma**, and every write is a server action checking session, Zod validation, plan and rate limits with user-scoped queries.
- Added **OAuth authentication** with NextAuth (credentials + GitHub, email verification via Resend), **Stripe** Free/Pro plans, file uploads to Cloudflare R2, and per-user **Redis rate limiting** via Upstash.
- Added **LLM-powered features** (auto-tagging, description generation, code explanation, prompt optimization) via an OpenAI-compatible API, plus JSON/ZIP import-export and a Vitest suite; auto-deployed on Vercel.
- Built a **Chrome / Edge extension** that saves selected text from any page as an item with AI tag suggestions, backed by a **REST API** authenticated with hashed, revocable personal access tokens.

TeleMetrix – real-time Formula Student telemetry dashboard

[GitHub](#)

Processing (Java), Arduino, ESP32, XBee Pro S2C, Serial/UART, CSV logging

- Built a full-HD track-side dashboard parsing a **real-time serial stream** at 230,400 baud into live gauges, bars and status lights, validating every frame’s field count so **malformed packets never reach the display**.
- Added rolling time-series traces on **sliding-window buffers** and a G-G friction-circle plot beside a live FPV feed, plus an Arduino signal simulator for bench testing without the car.

DAQ – data acquisition & driver dashboard, Team Fateh

[GitHub](#)

ESP32, FreeRTOS, C++ (Arduino), PlatformIO, CAN, SPI, I2C, UART, XBee, Nextion HMI, FastLED, MPU6050, KiCad

- Engineered an **ESP32 data acquisition system** across CAN, SPI, I2C and UART, decoding PE3 ECU **CAN frames at 1 Mbps** alongside brake pressure, gear and MPU6050 IMU data to capture the car’s full powertrain state.
- Architected **FreeRTOS task scheduling** on the dual-core ESP32 for a **4× performance boost** at a 30 Hz data rate, streaming live data over XBee at ~3 km range while logging to microSD over SPI.
- Built the Nextion HMI driver dashboard with a FastLED shift light, a **CAN watchdog**, RPM-based speed estimation and a moving-average filter for stable G-force readings.

Technical Skills

Languages	Java, Python, C/C++, TypeScript, JavaScript, SQL
Backend	Spring Boot, Spring Cloud, Spring AI, Django / DRF, Hibernate / JPA, Node.js, REST APIs
Frontend	React, Next.js, Tailwind CSS
Data & Messaging	PostgreSQL, pgvector, Redis, Apache Kafka, S3 Object Storage
AI / LLM	RAG, Hybrid Search, Embeddings, Spring AI, Gemini API, OpenAI API
Cloud & DevOps	Docker, Kubernetes, GitHub Actions (CI/CD), Git, Linux, AWS
Testing	JUnit, Testcontainers, Vitest, Playwright
Embedded & Robotics	ESP32, FreeRTOS, Arduino, CAN / SPI / I2C / UART, ROS 2, SLAM, Gazebo
Concepts	Distributed Systems, Event-Driven Architecture, System Design, Data Structures & Algorithms

Achievements & Awards

MITACS Globalink Research Internship

Summer 2024

Fully funded international research internship, University of Calgary, Canada

Academic Merit – Thapar Institute

2021 – 2025

<u>Gold Medal & Dean’s List</u>	1st rank in ECE Department (2021–22); led to branch upgrade to COE
<u>Merit Scholarship I</u>	Top 35 in college (2022–23)
<u>Merit Scholarship III</u>	Top 10% of batch (2023–24, 2024–25)
<u>Entry Merit Scholarship</u>	All India Rank, JEE Main 2021 (2021–22)

Formula Student with Team Fateh

2022 – 2025

<u>FS Italy’24</u>	5th (global)	4th: Skidpad, Acceleration, Autocross
<u>FS Bharat’24</u>	3rd	1st: Static Events, Engineering Design, Cost & Manufacturing
<u>SUPRA’23</u>	2nd	1st: Static & Dynamic Events, Engineering Design, Autocross

Positions of Responsibility

Electrical & Electronics Head

Apr 2024 – Apr 2025

Team Fateh – Formula Student Team, TIET

FSAE

- **Led the Electrical & Electronics subsystem** through the team’s **combustion-to-electric powertrain transition**, co-designing the low-voltage architecture with pre-charge, discharge and safety circuits to meet Formula Student EV rules.

Electronics & Data Acquisition Engineer

Oct 2022 – Mar 2024

Team Fateh – Formula Student Team, TIET

FSAE

- Owned the car’s data acquisition, driver dashboard and **real-time telemetry**, building an **ESP32 + FreeRTOS DAQ** over CAN, SPI, I2C and UART that streamed live sensor data to track-side engineers for vehicle-dynamics validation.